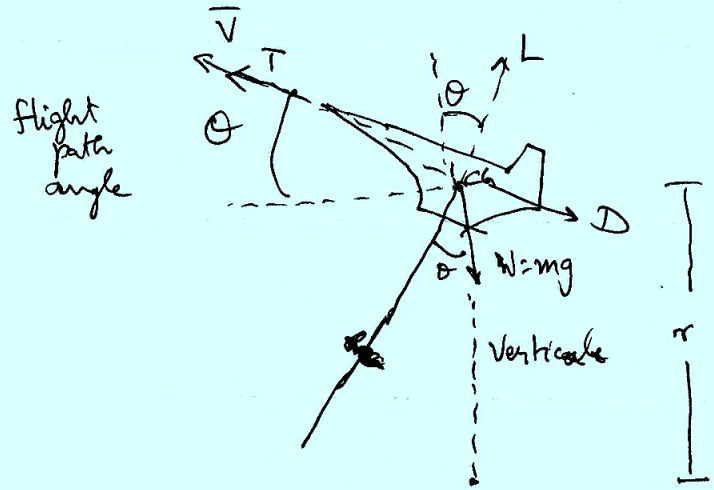
2/21/2019.

Consider a vehicle that takes off from a runway, flies thru the atmosphere, & achieves LEO ($M \sim 25$ reqd)

Begin w/ Newtons 2nd law



$$\Sigma F = m\frac{dV}{dt}$$

$$m\frac{dV}{dt} = \hat{\tau} - D - mg\sin\theta \qquad (\theta: \text{flight path angle})$$

Multiply by V

$$mV\frac{dV}{dt} = V(\hat{\tau} - D) - mgV\sin\theta$$

Let r: radial distance from the center of earth.

$$\frac{dr}{dt} = V\sin\theta$$

Can use chain rule to show

$$V\frac{dV}{dt} = \frac{d}{dt}\left(\frac{V^2}{2}\right)$$

$$m\frac{d}{dt}\left(\frac{V^2}{2}\right) = V(\hat{\tau} - D) - mg\frac{dr}{dt}$$

Recall thrust power $\tau V = \eta_o H_{vf} \dot{m}_f$

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& $\tau^{n} = \tau - D = \tau(1-\phi_e)$

$$m\frac{d}{dt}\left(\frac{v^2}{2}\right) = v\left(\tau - D_e - D\right) - mg\frac{dr}{dt}$$

$$m\frac{d}{dt}\left(\frac{v^2}{2}\right) = \tau v\left(1 - \frac{(D_e+D)}{\tau}\right) - mg\frac{dr}{dt}.$$

$$= \eta_0 H_{vf}\dot{m}_f\left(1 - \left(\frac{D_e+D}{\tau}\right)\right) - mg\frac{dr}{dt}.$$

Now assume that the mass of vehicle is slowly changing such that

$$\frac{dm}{dt} = -\dot{m}_f$$

$$m\frac{d}{dt}\left(\frac{v^2}{2}\right) = -\eta_0 H_{vf}\left(1 - \left(\frac{D_e+D}{\tau}\right)\right)\frac{dm}{dt} - mg\frac{dr}{dt}$$

Cancel out $\frac{d}{dt}$ terms

$$m\,d\left(\frac{v^2}{2}\right) = -dm\left\{\eta_0 H_{vf}\left(1 - \left(\frac{D_e+D}{\tau}\right)\right)\right\} - mg\,dr$$

Rearrange :

$$\boxed{\frac{dm}{m} = -\left\{\frac{d\left(\frac{v^2}{2}\right) + g\,dr}{\eta_0 H_{vf}\left(1 - \left(\frac{D_e+D}{\tau}\right)\right)}\right\}}$$

Transatmospheric flight

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$d\left(\frac{V^2}{2}\right)$: differential KE

$g\,dr$: differential PE.

→ So a change in mass dm due to ~~fuel~~ fuel results in a change in KE & PE!

→ Fuel is being converted into KE & PE by propulsion system.

→ Thus, the effective overall efficiency for transatmospheric flight is $\eta_0\left(1 - \left(\frac{D_e + D}{T}\right)\right)$

internal perf of engine

External performance of airframe + integration

IMPORTANT:

Note that this eqn doesn't work for cruise (@ cruise $d\left(\frac{V^2}{2}\right) = 0$; $g\,dr = 0$; $T = D_e + D$)

Again, just as we did for cruise, assume $\eta_0 H_{vf}\left(1 - \frac{(D_e + D)}{T}\right) = ¢$, so we can integrate to yield.

$$\ln\left(\frac{m_{final}}{m_i}\right) = -\left\{\frac{\left(\frac{V_f^2}{2} - \frac{V_i^2}{2}\right) + \int_0^r g\,dr}{\eta_0 H_{vf}\left(1 - \left(\frac{D_e + D}{T}\right)\right)}\right\}$$

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$$g = \frac{g_0 r_0^2}{r^2}$$

$$\int_{r_0}^{r} g\,dr = g_0 r_0 \left(1 - \frac{r_0}{r}\right)$$

$$\boxed{\Pi_f\Big|_A = \frac{m_f}{m_i} = 1 - \exp\left\{ \frac{-\left(V_f^2/2 - \frac{V_i^2}{2}\right) + g_0 r_0 \left(1 - \frac{r_0}{r_f}\right)}{\eta_0 H_{vf} \left(1 - \left(\frac{D_e + D}{\tau}\right)\right)} \right\}}$$

transatmospheric

Transatmospheric trajectory for $\eta_0 H_{vf} \left(1 - \left(\frac{D_e + D}{\tau}\right)\right)$

Recall for cruise

$$\Pi_f\Big|_c = 1 - \exp\left\{ \frac{-g_0 R}{\eta_0 H_{vf} (1 - \phi_e) \frac{C_L}{C_D}} \right\}$$

& effective overall efficiency for cruise was $\eta_0 (1 - \phi_e) \frac{C_L}{C_D}$

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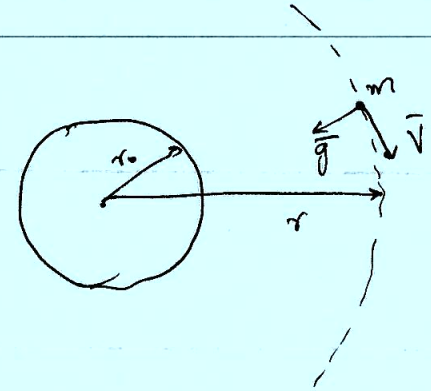
Orbital Energy



Consider a satellite of mass m in a circular orbit of radius r

Newton's 2nd law

$$\sum \bar{F} = m\frac{d\bar{V}}{dt}$$

$$= m\bar{g}$$

(no τ or D)

Centrifugal accel

$$m\frac{dV}{dt} = \frac{mV^2}{r} = mg$$

$$\Rightarrow V_{orb} = \sqrt{gr}$$

$$g = g_0\left(\frac{r_0^2}{r^2}\right)$$

$$\Rightarrow V_{orb} = \left(\frac{g_0 r_0^2}{r}\right)^{1/2}$$

Define orbital energy $\mathcal{E} = KE + PE$

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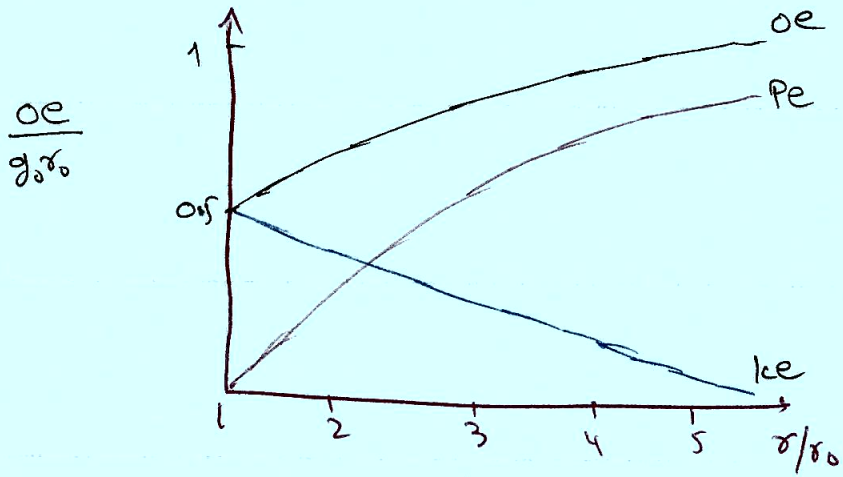
$$Oe = \frac{1}{2} V_{orb}^2 + \int g\,dr$$

$$= \frac{1}{2} g_0 \frac{r_0^2}{r} + g_0 r_0 \left(1 - \frac{r_0}{r}\right)$$

$$\boxed{\frac{Oe}{g_0 r_0} = 1 - \frac{1}{2}\frac{r_0}{r}}$$



@ LEO, $h \cong 300$ km $\quad r_0 = 6400$ km.

$$\frac{r}{r_0} = 1.05$$

ratio of $\frac{KE}{PE} = \dfrac{\frac{1}{2}\left(\frac{r_0}{r}\right)}{1 - r_0/r} \approx 12$

$$V_{orbit} = \sqrt{rg} = \sqrt{g_0 r_0^2/r} = \sqrt{g_0 r_0 \left(1/(r/r_0)\right)}$$

$$\boxed{V_{orb} = 7.7 \text{ km/s}}$$

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Orbital Period : $OP = \frac{2\pi r}{V_{orb}} = 2\pi \left(\frac{r^3}{g_0 r_0^2}\right)^{1/2}$

$= 90 \text{ min}$

@ GEO (geosynchronous orbit)

Orbital Period = 24 hrs.

$$r = \left(\frac{(OP)^2}{(2\pi)^2} \cdot g_0 r_0^2 \right)^{1/3} = 42000 \text{ km}$$

$$\frac{r}{r_0} = 6.6.$$

Vel. @ GEO $\quad V_{orb,geo} \simeq 307 \text{ km/s}$

ratio of $KE/PE = 0.09.$

Recall for transatmospheric flight

$$\Pi_f = 1 - \exp\left\{ - \frac{\left(v_f^2/2 - v_i^2/2\right) + g_0 r_0 \left(1 - \frac{r_0}{r}\right)}{\eta_0 H_{vf} \left(1 - \left(\frac{D + D_e}{\tau}\right)\right)} \right\}$$

Using out concept of oe & neglecting v_i

$$\Pi_f = 1 - \exp\left\{ \frac{-g_0 r_0 \left(1 - \frac{1}{2}\frac{r_0}{r}\right)}{\eta_0 H_{vf} \left(1 - \left(\frac{D + D_e}{\tau}\right)\right)} \right\}$$

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The lower the Π_f, the higher the overall efficiency η_o to achieve a given objective.

Empty mass fraction

Recall $m_i = m_e + m_p + m_f$

$$\Pi_f = \frac{m_f}{m_i}$$

Empty mass fraction

$$\Pi_e = \frac{m_e}{m_i}$$

For general we want high Π_f & low Π_e. But Π_e is difficult to determine in the design stage of a vehicle. Π_e is function of initial mass, & decreases with m_i

From historical data of high performance aircraft

$$\Pi_e = 2.34\, m_i^{-0.13}$$

We assume this holds for hypersonic a/c

Payload mass ratio

$$\Pi_p = \frac{m_p}{m_i}$$

Payload is why we are building the a/c, so want this to be large.

$$\Pi_f + \Pi_e + \Pi_p = 1$$

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Note : the text uses initial mass ratio $\frac{m_i}{m_p} \equiv \Gamma_m$

& $\Gamma_m = \Pi_p^{-1} = \left(1 - \Pi_f - \Pi_e\right]^{-1}$

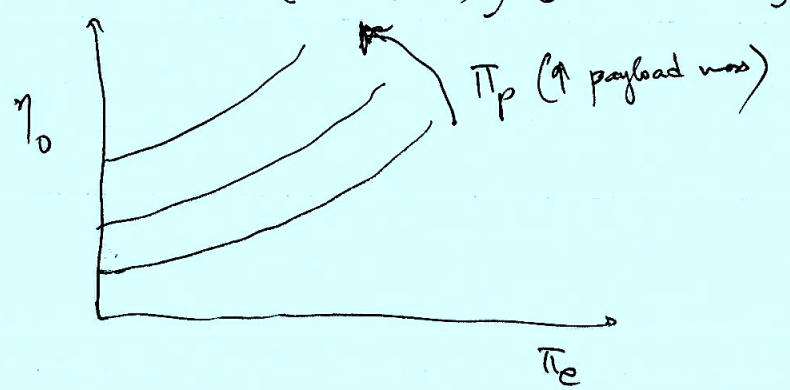
So we want to minimize Γ_m

Can write overall efficiency in terms of Π_i, Π_m

For cruise, $$\eta_o = \frac{g_o R}{H_{vf}\left(1-\phi_e\right)\frac{C_L}{C_D}\left(1 - \ln\left(\Pi_e + \Pi_p\right)\right)}$$

For transatm. $$\eta_o = \frac{g_o r_o \left(1 - \frac{1}{2}\frac{r_o}{r}\right)}{H_{vf}\left(1 - \left(\frac{D_e + D}{\tau}\right)\right)\left(1 - \ln\left(\Pi_e + \Pi_p\right)\right)}$$

For cruise



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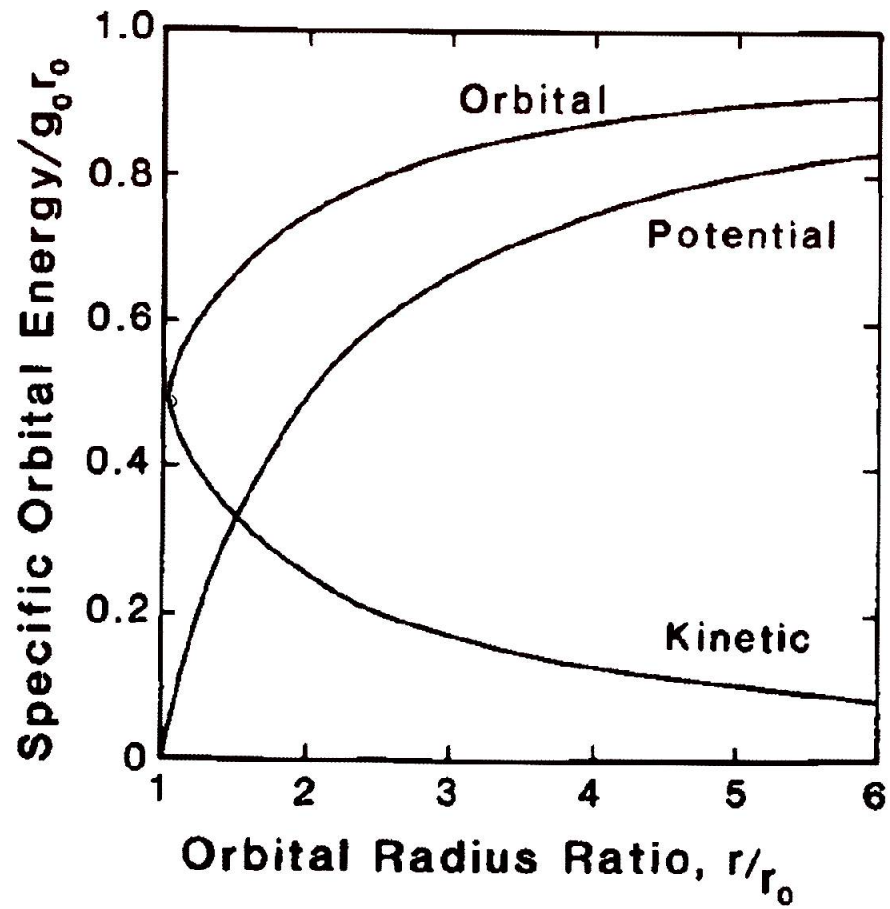


Fig. 3.6 Specific orbital energy, specific kinetic energy

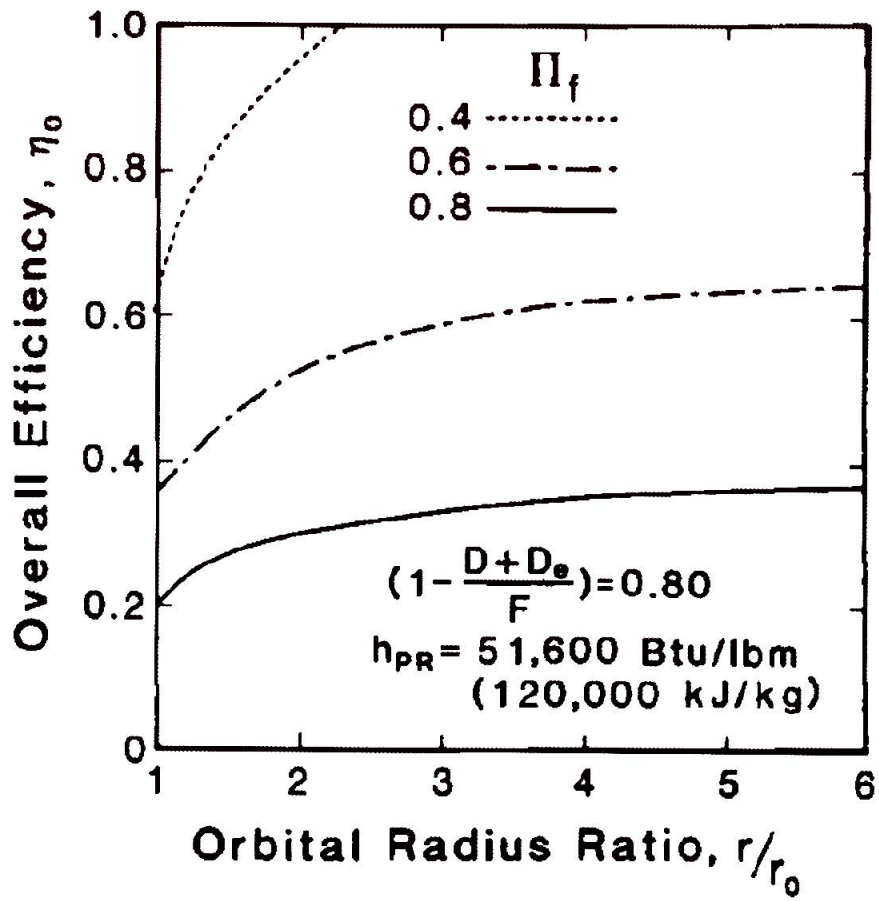


Fig. 3.7 Required airbreathing engine overall efficienc

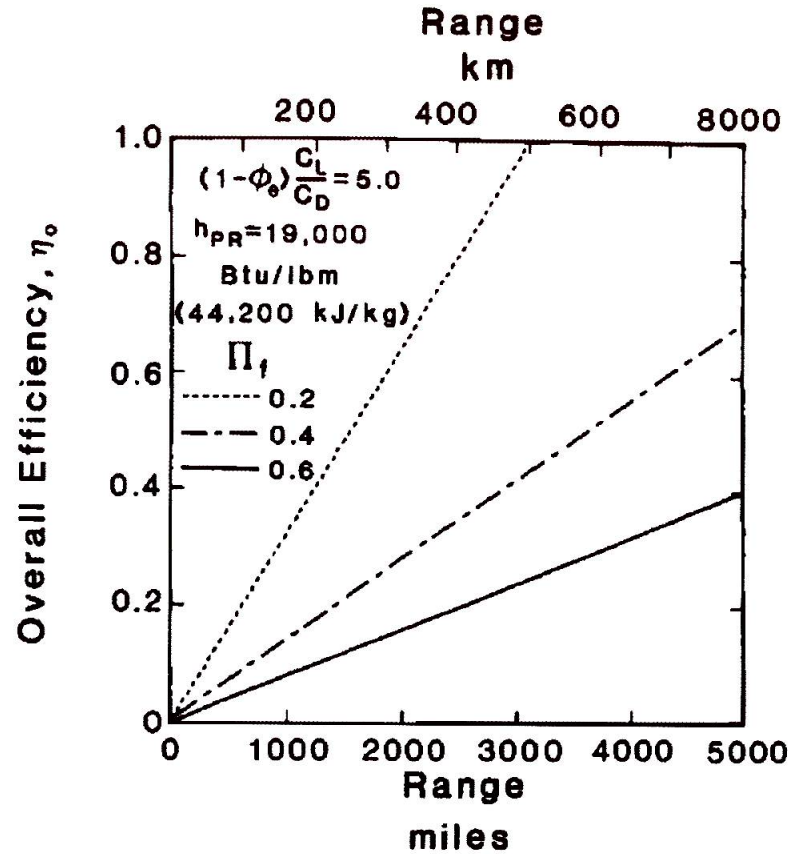


Fig. 3.3 Required airbreathing engine overall efficiency as a

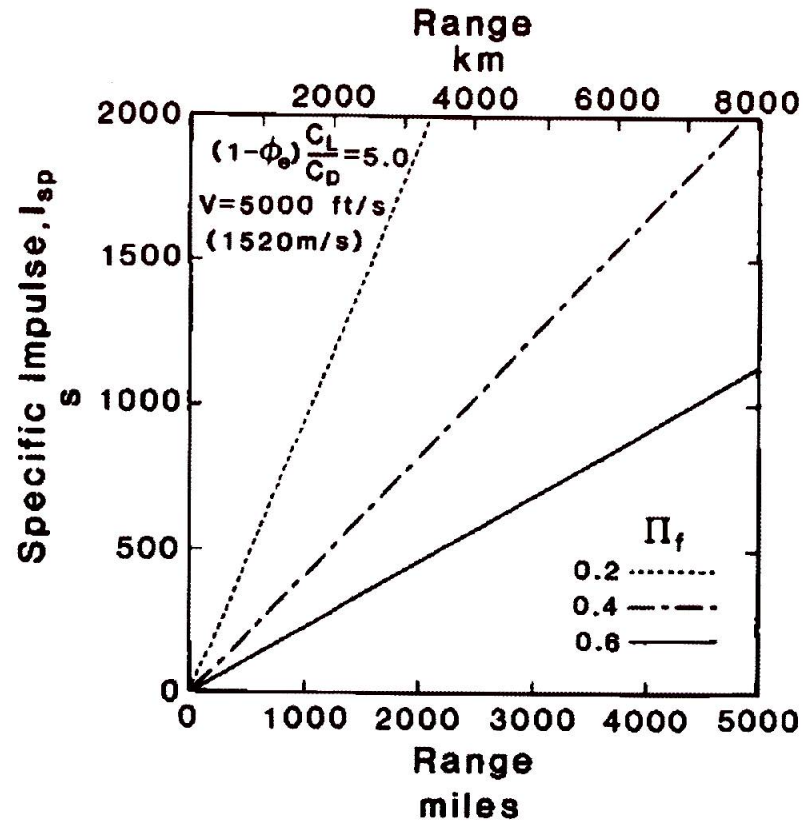
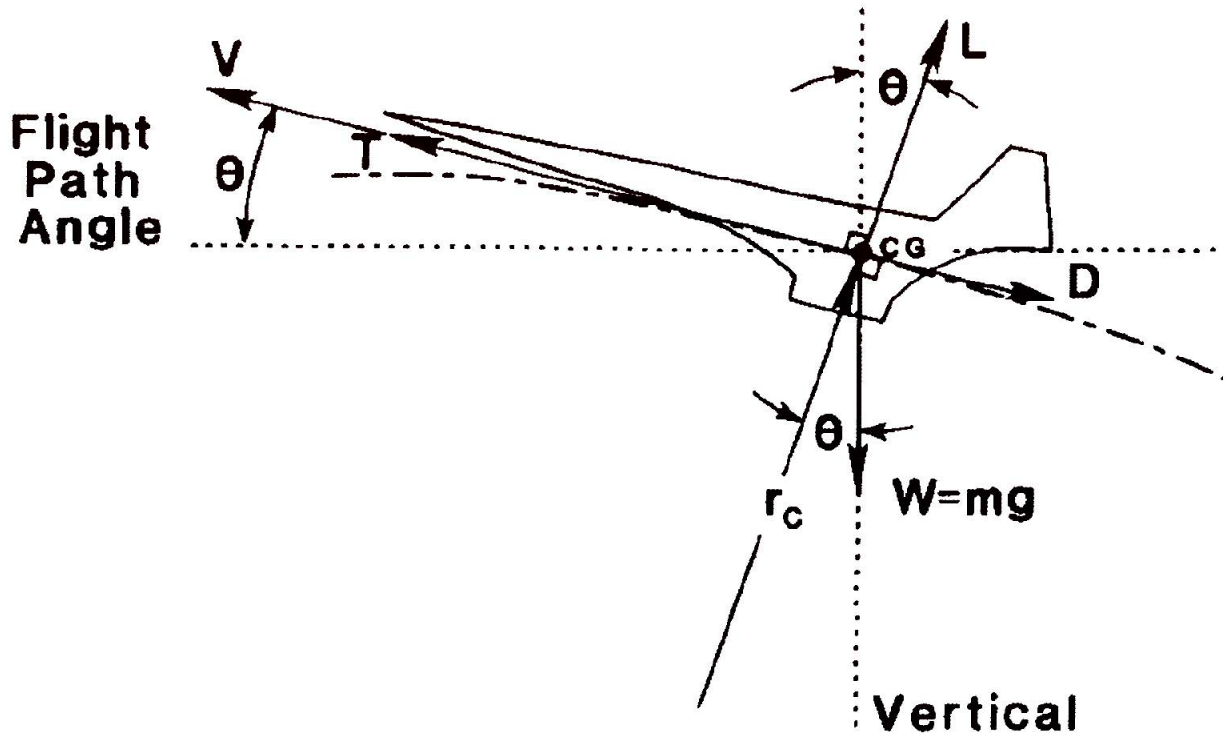


Fig. 3.4 Required rocket engine specific impulse as a functi
hypersonic aircraft cruise range and fuel (plus oxidizer) mass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Flight
Path
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